# RAG Evaluation Metrics and System Architecture

**System Design & Formulations**

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## 1. System Architecture Diagram

Below is the schematic representation of the Retrieval-Augmented Generation (RAG) pipeline showing the query $Q$, document database $D$, retrieved contexts $(C_1, C_2, C_3)$, and generated answer $A$.

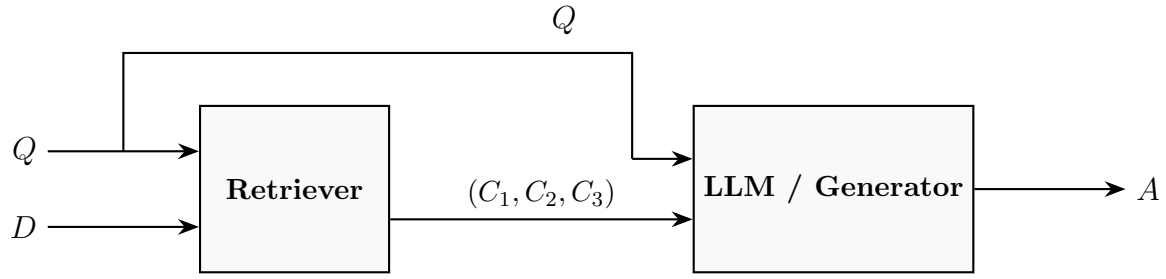



Figure 1: Retrieval-Augmented Generation (RAG) Data Flow Architecture

## 2. Mathematical Formulations of RAG Metrics

### 2.1. Faithfulness

$$\text{Faithfulness}(A, C_{1:n}) = \frac{\text{Number of claims in } A \text{ that can be inferred from the context } C_{1:n}}{\text{Total number of claims in } A}$$

### 2.2. Context Relevance

$$\text{ContextRelevance}(Q, C_{1:n}) = \frac{1}{n}\sum_{i=1}^{n}\left(\frac{Q \cdot C_i + 1}{2}\right) = \frac{1}{2}\left(\frac{1}{n}\left(Q \cdot \sum_{i=1}^{n} C_i\right) + 1\right)$$

$$\text{ContextRelevance}(Q, C_{1:n}, \beta) = \frac{\sum_{i=1}^{n} \mathbb{I}(Q \cdot C_i > \beta)}{n} \quad \text{where } \beta \text{ is the bound on the semantic similarity}$$

### 2.3. Answer Relevance

$$\text{AnswerRelevance}(Q, A) = \frac{1}{n}\sum_{i=1}^{n}\left(\frac{Q \cdot Q_i(A) + 1}{2}\right) = \frac{1}{2}\left(\frac{1}{n}\left(Q \cdot \sum_{i=1}^{n} Q_i(A)\right) + 1\right)$$

### 2.4. Context Recall

$$\text{ContextRecall}(GT, C_{1:n}) = \frac{1}{n}\sum_{i=1}^{n}\left(\frac{GT \cdot C_i + 1}{2}\right) = \frac{1}{2}\left(\frac{1}{n}\left(GT \cdot \sum_{i=1}^{n} C_i\right) + 1\right)$$